

Chapter 1: Linear Regression & Logistic Models

One concept, one exercise per slide — practice as you go

Concept: What Regression Is Trying to Do

- Description: summarize how y relates to x in the data you have
- Prediction: guess y for a new observation's x
- Explanation: estimate how much x influences y , holding other things fixed
- Causal inference: what happens to y if we intervene and change x ?



Exercise: A retailer fits a model of sales vs. advertising spend to decide whether to increase next quarter's ad budget. Which of the four goals above does this require — and is a plain regression coefficient enough to justify the decision?

Concept: Slope and Intercept

- A line is written $y = a + b \cdot x$
- a (intercept): the value of y when $x = 0$
- b (slope): how much y changes per one-unit increase in x
- Rearranged: $x = (y - a) / b$



Exercise: A line has intercept $a = 5$ and slope $b = -2$. What is y when $x = 3$? What value of x gives $y = 1$?

Concept: Mean, Variance, and Covariance

- Mean: add the values, divide by the count
- Variance: how spread out values are around the mean
- Covariance: whether two variables move together
- The simple regression slope is $\text{Cov}(x,y) / \text{Var}(x)$



Exercise: For $x = [1, 2, 3]$ and $y = [2, 4, 6]$, compute $\bar{x}\bar{y}$, and the slope of the regression line using $\text{slope} = \text{Cov}(x,y)/\text{Var}(x)$.

Concept: Least Squares and Residuals

- Residual: $e = y - \hat{y}$ (observed minus predicted)
- OLS chooses the line minimizing the sum of squared residuals $\sum e^2$
- Squaring penalizes large mistakes more and has clean calculus
- ε (true disturbance) $\neq$ e (observed residual) — e is only an estimate of ε



Exercise: A fitted line gives $\hat{y} = 4.6$ at $x = 4$. The observed value is $y = 4$. What is the residual? Is the model over- or under-predicting at this point?

Concept: Multiple Regression and the Design Matrix

- $y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \varepsilon$ — many predictors, one model
- Stack predictors into a design matrix X ; solve $\hat{\beta} = (X^T X)^{-1} X^T y$
- Each β_j is interpreted "holding all other predictors constant"



Exercise: A model predicts house price from square footage (x_1) and number of bedrooms (x_2), giving $\beta_1 = 150$ and $\beta_2 = 5000$.
What does $\beta_1 = 150$ mean in plain language?

Concept: The Assumptions Behind OLS

- Linearity, independence, no perfect multicollinearity
- Zero conditional mean: $E[\varepsilon | X] = 0$ — no systematic error given predictors
- Homoskedasticity: constant error variance across X
- Normality: needed for exact small-sample inference



Exercise: A residual-vs-fitted plot shows a clear funnel shape (residuals fan out as fitted values increase). Which assumption is violated, and which part of the analysis does it put at risk — the coefficients themselves, or the standard errors?

Concept: Standard Errors, Confidence Intervals, p-values

- A coefficient's standard error measures how much it would vary across fresh samples
- A 95% CI is a range of plausible values built from the estimate $\pm$ SE
- A p-value asks: how surprising is this data if the true effect were zero?
- Statistical significance $\neq$ practical importance



Exercise: A coefficient estimate is 0.8 with a 95% CI of [0.75, 0.85]. A second coefficient is 12.0 with a 95% CI of [-3.0, 27.0]. Which estimate is more precise, and which is more likely to be practically important if both are statistically significant?

Concept: R^2 and Measuring Fit

- $R^2 = 1 - (SS_{\text{residual}} / SS_{\text{total}})$: proportion of variance explained
- $R^2 = 0$ means no better than predicting the mean every time
- Adding any predictor can only increase raw R^2 — use adjusted R^2 to penalize this
- Low R^2 does not mean a coefficient is unimportant



Exercise: A model has $SS_{\text{total}} = 500$ and $SS_{\text{residual}} = 100$. Compute R^2 . If a second predictor is added and SS_{residual} drops to 95, did the model necessarily improve in a meaningful way?

Concept: Residual Diagnostics

- Residuals vs. fitted: look for curves (non-linearity) or funnels (heteroskedasticity)
- Q-Q plot: checks whether residuals are normally distributed
- Leverage and Cook's distance: flag unusual or influential points
- A model that runs without error can still badly violate its assumptions



Exercise: Anscombe's Quartet is four datasets with identical means, variances, correlation, and regression line — but very different shapes. What single diagnostic step would reveal the difference that summary statistics hide?

Concept: Multicollinearity

- Highly correlated predictors make individual coefficients unstable
- Predicted values usually remain reliable even when coefficients are shaky
- VIF (Variance Inflation Factor) quantifies the instability
- VIF above roughly 5–10 is commonly flagged as concerning



Exercise: Two predictors — height in centimeters and height in inches — are both included in a regression. Predict what will happen to their individual coefficients and standard errors, and explain why the model's overall predictions are still likely to be fine.

Concept: Ridge and Lasso Regularization

- Ridge adds a penalty $\lambda \sum \beta_j^2$ — shrinks coefficients toward zero, keeps them all
- Lasso adds a penalty $\lambda \sum |\beta_j|$ — can shrink coefficients to exactly zero
- Regularization requires standardized features first
- More regularization = more bias, less variance



Exercise: You have 100 candidate predictors and suspect only 5 genuinely matter. Would ridge or lasso be more useful for identifying which 5? Why?

Concept: Logistic Regression and Log-Odds

- For binary outcomes, model the log-odds: $\text{logit}(p) = \log(p/(1-p)) = \beta_0 + \beta_1 x$
- Invert to get a probability: $p = 1 / (1 + \exp(-(\beta_0 + \beta_1 x)))$
- $\exp(\beta_j)$ is the multiplicative change in ODDS, not probability, per unit of x



Exercise: A probability of an event is $p = 0.8$. Compute the odds. Then, if a coefficient's $\exp(\beta) = 1.5$, what happens to the odds (not the probability) when x increases by 1?

Concept: Evaluating Classification Models

- Precision: of predicted positives, how many were correct?
- Recall: of actual positives, how many did we find?
- Accuracy is misleading when classes are imbalanced
- ROC-AUC and PR-AUC summarize performance across all thresholds



Exercise: Out of 100 emails, a spam filter flags 20 as spam. Of those, 15 are genuinely spam. There were 25 genuine spam emails total. Compute precision and recall.

Concept: Cross-Validation and Data Leakage

- Train/test split: hold out data to estimate out-of-sample performance
- K-fold cross-validation: repeat the split k times for a more stable estimate
- Leakage: letting test-set information influence training
- Fit scaling/imputation only on the training fold — never on the full dataset first



Exercise: A colleague standardizes the entire dataset (mean/variance from all rows) before doing a train/test split, then reports high test accuracy. What is wrong with this workflow?

Concept: From-Scratch vs. Library Implementations

- Pure Python: every calculation explicit — nothing is a black box
- NumPy: vectorized linear algebra underneath almost everything else
- statsmodels: rich inference — coefficients, SEs, p-values, diagnostics
- scikit-learn: fast, consistent pipelines for prediction



Exercise: You need to report a coefficient's p-value and confidence interval for a research paper. Which library is the right tool, and why not the other one?

Concept: Common Mistakes

- "Significant" $\neq$ "important" — always report magnitude and units
- Scaling before splitting causes leakage
- Reusing the test set repeatedly turns it into training information
- Deleting every outlier without investigating first



Exercise: A model reports $p < 0.001$ for a coefficient of 0.0003 on an outcome measured in millions of dollars. Is this result practically important? What else would you want to know?

Concept: Causal Caution — Correlation Is Not Causation

- A confounder is an unmeasured factor driving both x and y
- Omitted-variable bias: leaving out a confounder correlated with x biases $\hat{\beta}$
- High predictive accuracy does not imply a safe intervention
- Causal claims need a causal design, not just a regression coefficient



Exercise: Ice cream sales strongly predict drowning deaths across months. Identify the confounder, and explain why reducing ice cream sales would not reduce drownings.

Mastery Check: Are You Ready to Move On?

If you can answer every question below confidently, you have mastered Chapter 1.

1. Explain the difference between epsilon (the true disturbance) and e (the observed residual).
2. Derive the simple regression slope $b_1 = S_{xy}/S_{xx}$ from the normal equations.
3. State all five core OLS assumptions and name which guarantee each one supports.
4. Explain why a prediction interval is always wider than a confidence interval for the mean.
5. Give an example where R^2 is low but a coefficient is still substantively important.
6. Explain why near multicollinearity damages coefficient interpretation but not necessarily prediction.
7. Explain why ridge requires feature scaling and why lasso can produce exactly-zero coefficients.
8. Convert a probability to odds and to log-odds, and explain what $\exp(\beta)$ means in logistic regression.
9. Explain why accuracy is a misleading metric under severe class imbalance.
10. Explain why fitting a scaler on the full dataset before splitting causes data leakage.
11. Implement simple linear regression from scratch in pure Python and verify it against a known example.
12. Fit the same model in statsmodels and scikit-learn and explain why you'd choose one over the other.
13. Give a concrete example where a regression shows strong association but no plausible causal path.
14. Explain, in one sentence each, what a p-value is and what it is NOT.